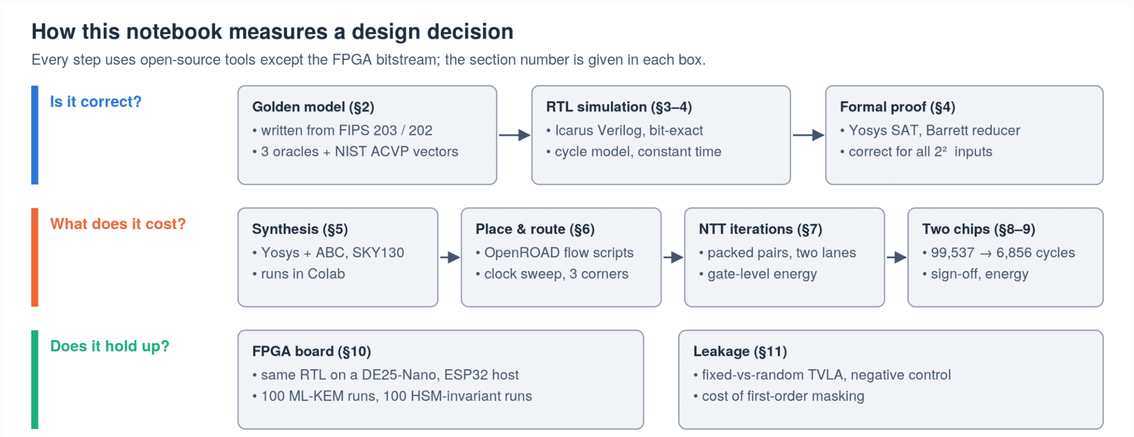
Measuring the Design Decisions of an Open-Source Post-Quantum HSM Chip

ML-KEM-512 NTT and Keccak, from a Python golden model to a SKY130 layout

Nguyen Tan Dat · University of Science, VNU-HCM (HCMUS) · IEEE SSCS Code-a-Chip, ISSCC 2027

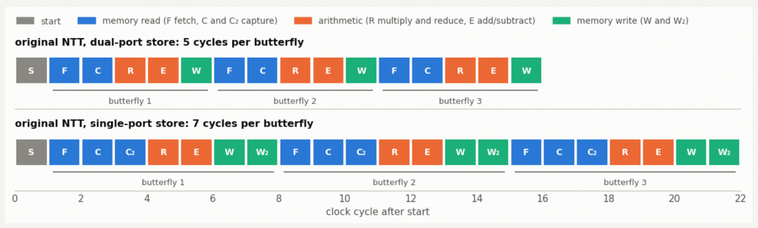
run it in Colab

What did each architectural decision made for the ASIC actually cost in area, latency and energy, what does the design still leak, and can anyone re-derive those numbers with open tools?

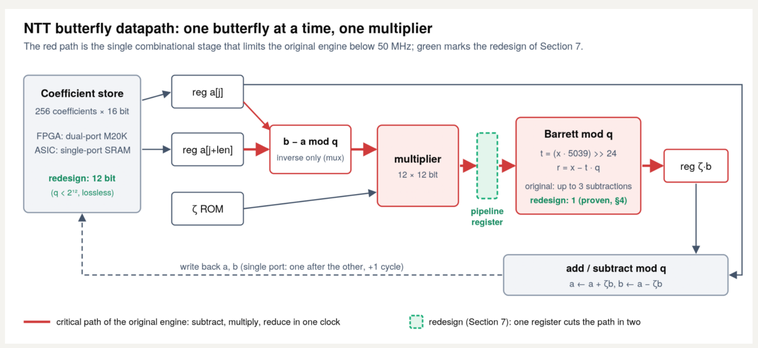


I. Is it correct?

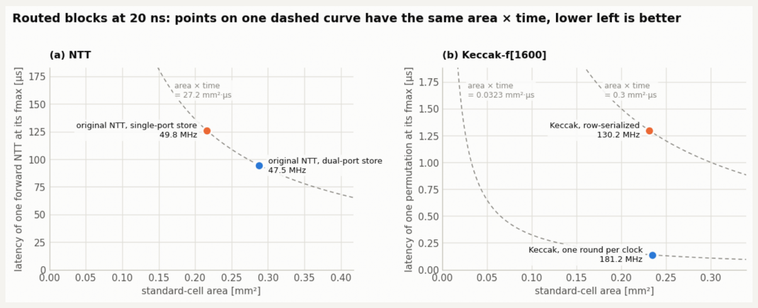
- 25/25 NIST ACVP key generations reproduced byte for byte by the complete RTL
- 0 mismatches over 208,896 NTT coefficients and 404 Keccak permutations, at a constant latency
- The Barrett reducer is proven correct for all 2^{24} inputs, and one subtraction suffices
- The routed netlists reproduce the golden results at gate level



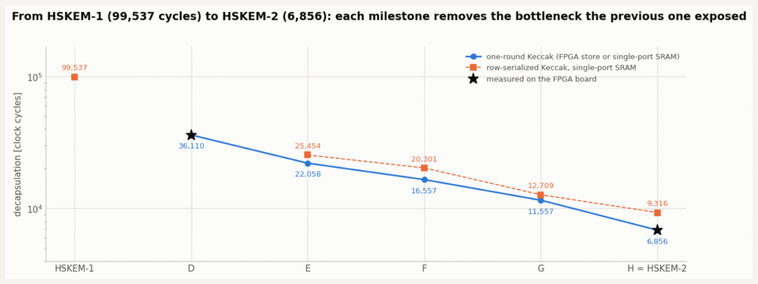
The controller visits the 896 butterflies in the order of FIPS 203, Algorithm 9; the single-port store needs two extra cycles per butterfly.



II. What does it cost?

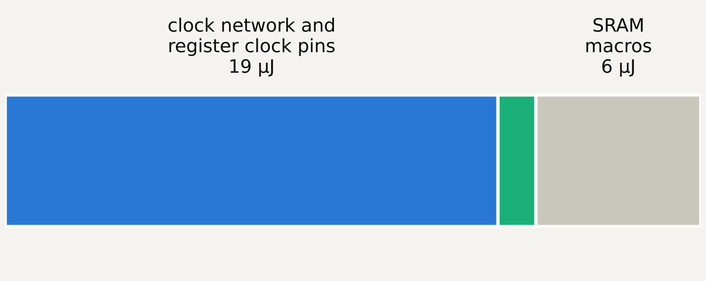


A first measured iteration of the NTT, after routing: fmax 50 → 83 MHz, area -21%, area × time -46%.



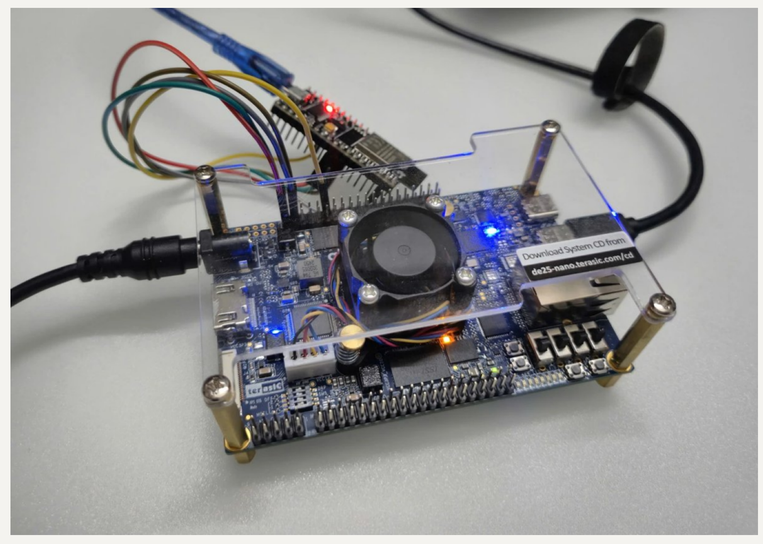
A second iteration stores two coefficients in every SRAM word: a forward NTT takes 14 instead of 130 µs and 0.16 instead of 0.50 µJ. With a two-lane engine and a streamed datapath that removes no check, a decapsulation takes 6,856 instead of 99,537 cycles (HSKEM-1 → HSKEM-2), as the board confirms.

A decapsulation on HSKEM-2: 27 µJ

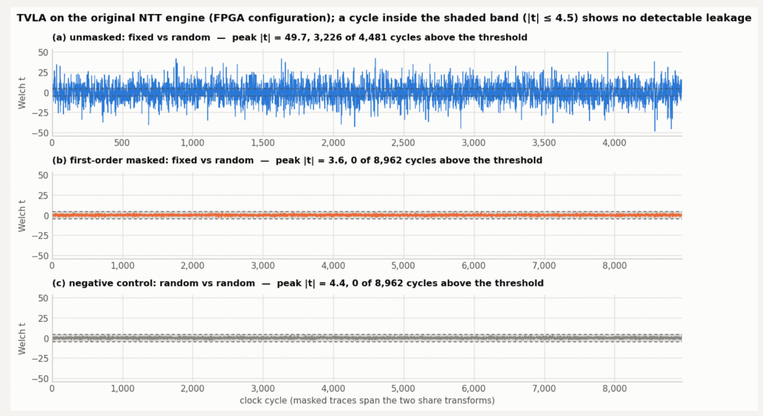


HSKEM-2, the signed-off SKY130 chip of the streamed system, spends 27 µJ per decapsulation instead of HSKEM-1's 0.34 mJ. The rest is set by the integration: gating the clock of idle blocks and the chip selects of the macros would save roughly 32 to 48 %.

III. Does it hold up?



The same RTL on the DE25-Nano: 100/100 two-role ML-KEM runs pass.



Simulated TVLA: largest |t| 49.7 unmasked, 3.6 with first-order masking (threshold 4.5).

What transfers

- Weigh a block by how often the system waits for it.
- Measure the integrated chip, not only its blocks.
- Check the check: eleven results here first looked right and were not.
- Tie the prose to the data: assertions stop the notebook when they part.